Chapter 10
Energy Methods

In this chapter, we will learn a new topic *Energy Methods* which is an alternate method to obtain deformation of general structures including beams. Unlike *stress equilibrium equations* solving which we obtain displacement of every point in a body simultaneously, the *energy methods* gives displacement of only few special points in the body. However, as we will see, the new approach is algebraic in nature and much simpler to solve than partial differential equations. We will first learn about various concepts needed for understanding *energy methods*. We will then see how one can apply this method for solving beam deformation problems.

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10.1 Linearity and superposition of elasticity equations: NPTEL link

Consider an arbitrary body as shown in Figure 10.1.

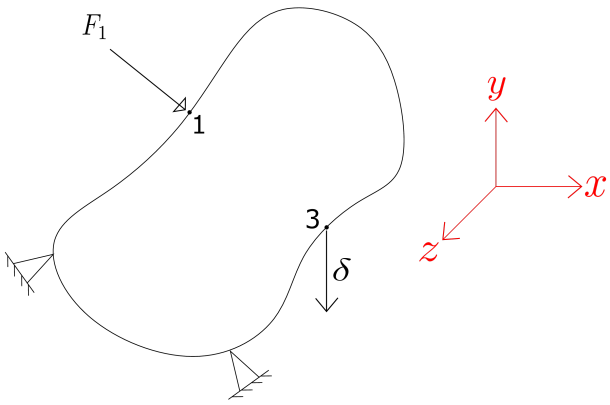


Figure 10.1: An arbitrary body clamped at two points and subjected to a point load elsewhere

It is clamped at two points and subjected to a concentrated load F_1 at point 1. Let us say we want to measure the deflection δ at a different point (say point 3 in the figure) which can be either on the surface of the body or within the body. How is the deflection δ related to force F_1? To explore it, let us look at the stress equilibrium equations as given below:

$$\begin{aligned}
\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} + b_x = 0, \\
\frac{\partial \tau_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} + b_y = 0, \\
\frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \tau_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + b_z = 0.
\end{aligned} \tag{1}$$

These equations are supplemented with the following boundary conditions:

$$\begin{aligned} \underline{\underline{\sigma}}\,\underline{n} &= \underline{t}_0 \quad \text{(traction boundary condition)}, \\ \underline{u} &= \underline{u}_0 \quad \text{(displacement boundary condition)}. \end{aligned} \tag{2}$$

Notice that the above equations are linear in stress components which, in turn, are linear in strain components (due to linear stress-strain relation). Finally, strain components are also linear in displacement components.[*] Thus, the governing equations become linear in unknown displacement components. Similarly, the boundary conditions are also linear in unknown displacement components. Due to this linearity of governing equations, the displacement of the body will be linearly related to externally applied load. In fact, this linearity also allows superposition principle to hold. The external load here consists of the body force and boundary traction/point loads. Coming back to the problem shown in Figure 10.1, δ_3 will thus be linearly related to the applied load F_1, i.e.,

$$\delta_3 \propto F_1 \text{ or } \delta_3 = k_{31} F_1. \tag{3}$$

The proportionality constant k_{31} is also called *influence coefficient*. If we change the point of location of force F_1, k_{31} will change. Similarly, if we change the point at which we are measuring the displacement, k_{31} will change. Hence, the influence coefficient depends on the location of the applied force as well as on the location of the point where displacement is being measured. In fact, it also depends on the direction of applied force and the direction in which displacement is being measured but does not depend on the magnitude of force. For example, displacement being a vector quantity, $\underline{\delta}_3$ will have three components δ_3^x, δ_3^y and δ_3^z. For each of the components, we will have different influence coefficients relating to each component of force applied at point 1, i.e.,

$$\begin{aligned} \delta_3^x &= k_{31}^{xx} F_1^x, \quad \delta_3^x = k_{31}^{xy} F_1^y, \quad \delta_3^x = k_{31}^{xz} F_1^z, \\ \delta_3^y &= k_{31}^{yx} F_1^x, \quad \delta_3^y = k_{31}^{yy} F_1^y, \quad \delta_3^y = k_{31}^{yz} F_1^z, \\ \delta_3^z &= k_{31}^{zx} F_1^x, \quad \delta_3^z = k_{31}^{zy} F_1^y, \quad \delta_3^z = k_{31}^{zz} F_1^z. \end{aligned} \tag{4}$$

Each of the equations above resembles Hooke's law for a spring where force in the spring is proportional to the elongation in the spring. The above equations hold only if just one (but any) of the components of force is acting at a time. In case all components are present simultaneously, one can apply the principle of superposition as mentioned earlier which would yield

$$\begin{aligned} \delta_3^x &= k_{31}^{xx} F_1^x + k_{31}^{xy} F_1^y + k_{31}^{xz} F_1^z, \\ \delta_3^y &= k_{31}^{yx} F_1^x + k_{31}^{yy} F_1^y + k_{31}^{yz} F_1^z, \\ \delta_3^z &= k_{31}^{zx} F_1^x + k_{31}^{zy} F_1^y + k_{31}^{zz} F_1^z. \end{aligned} \tag{5}$$

Note that the influence coefficients are unaffected whether only one component of force is present or all the components are present. The principle of superposition would hold similarly even if the force is acting at two (or more) different points in the body. For example, suppose we only apply a force F_1 at point 1 and no other force, the displacement at point 3 in a prescribed direction will be given by

$$\delta_3 = k_{31} F_1. \tag{6}$$

In another situation, if we apply a force F_2 at point 2 and no other force, the displacement at point 3 in the same direction will be given by

$$\delta_3 = k_{32} F_2. \tag{7}$$

In a third situation, if we apply force F_1 at point 1 and force F_2 at point 2 together, the displacement at point 3 (again in the same direction) will be

$$\delta_3 = k_{31} F_1 + k_{32} F_2. \tag{8}$$

Notice that the influence coefficients are the same as in the case when only one of the forces were acting.

An alternate way to prove superposition: NPTEL link

Consider an arbitrary body clamped at two locations. We apply a force F_1 at point 1 and measure vertical component of displacement at point 3 (see Figure 10.2a).

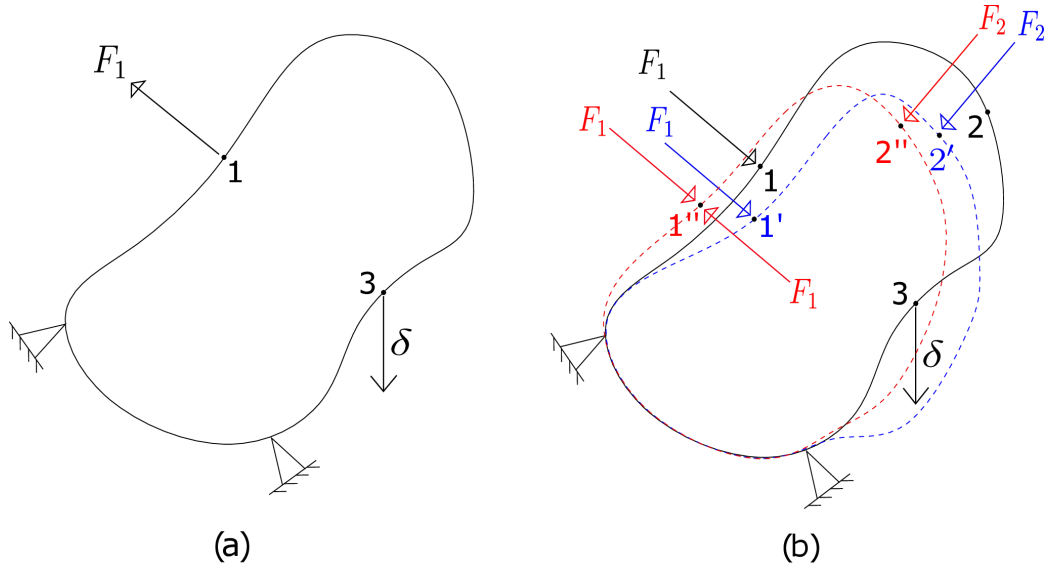


Figure 10.2: (a) A force F_1 applied on a body at a point 1 and vertical displacement measured at point 3 (b) A force F_2 applied on the deformed body (shown as blue dotted line) at point 2' (originally at 2)

The displacement will be

$$\delta_3 = k_{31}F_1. \tag{9}$$

Due to the application of force, the system must have deformed and changed its shape and size (see blue dotted lines in Figure 10.2b). In the next step, we apply force F_2 at point 2' (originally at 2) in the deformed body. So, both F_1 and F_2 are acting now. As F_2 is applied after the body has deformed, the influence coefficient k_{32} may have changed to k'_{32} which implies

$$\delta_3 = k_{31}F_1 + k'_{32}F_2. \tag{10}$$

In the third step, we apply a force $-F_1$ at point 1'' (originally at 1) in the current deformed body (see Figure 10.2b). Due to this, the total force acting at point 1 becomes zero and we are only left with a force F_2 at point 2'' (originally at 2). As the condition of the body has changed, the influence coefficient k_{31} may have changed to k''_{31}. So, the total vertical displacement at the same point 3 will be

$$\delta_3 = k_{31}F_1 + k'_{32}F_2 + k''_{31}(-F_1). \tag{11}$$

Alternately, as in the final configuration, we only have a force F_2 at point 2, the vertical displacement of point 3 should be

$$\delta_3 = k_{32}F_2. \tag{12}$$

Notice that we have used the original influence coefficient in the above equation as if only force F_2 were acting right from the beginning. We could say so because the displacement in an elastic body only depends on the final state of applied load and does not depend on how those loads were applied. Equating equations (11) and (12) since the final state of loading is the same, we obtain

$$\begin{aligned} &k_{31}F_1 + k'_{32}F_2 + k''_{31}(-F_1) = k_{32}F_2 \\ \Rightarrow\ &(k_{31} - k''_{31})F_1 + (k_{32} - k'_{32})F_2 = 0. \end{aligned} \tag{13}$$

Since this has to hold for arbitrary magnitudes of (F_1, F_2), their coefficients must vanish. Thus, we get

$$\begin{aligned} k_{31} &= k''_{31}, \\ k_{32} &= k'_{32}. \end{aligned} \tag{14}$$

This means that in step 2 when both F_1 and F_2 act, the displacement would simply be

$$\delta_3 = k_{31}F_1 + k_{32}F_2 \tag{15}$$

which proves that one can apply superposition principle.

Corresponding displacement/Work absorbing displacement

The displacement of the body at the point of application of force and in the same direction as the applied force is called the corresponding displacement. As this component of displacement is also responsible for the actual work done by the force, it is therefore also called the work absorbing displacement. Suppose we apply force at point 1 and measure the corresponding displacement δ_1, then

$$\delta_1 \propto F_1 \text{ or } \delta_1 = k_{11}F_1. \tag{16}$$

10.2 Energy stored in the body due to a applied force: NPTEL link

We now want to find the energy stored in a deformable body due to a force, say F_1. Would it simply be $F_1\delta_1$ where δ_1 is the corresponding displacement. To find an answer to this, consider a spring having spring constant k and clamped at one end. A force F is applied at its other end as shown in Figure 10.3a.

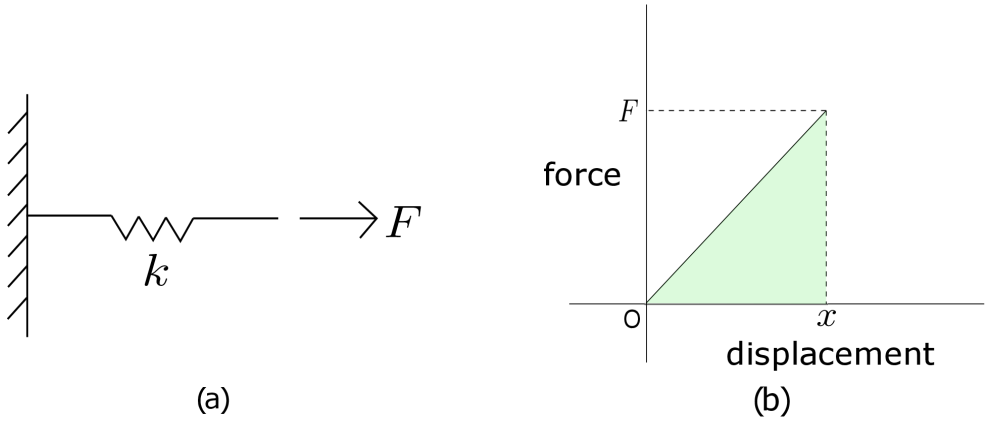


Figure 10.3: (a) A spring clamped at one end is stretched by a force F (b) Force-displacement curve for a spring that is stretched quasi-statically from 0 to x

We know that the energy stored in the spring due to this force is

$$\text{Energy stored} = \frac{1}{2}kx^2 = \frac{1}{2}kx \cdot x = \frac{1}{2}F \cdot x. \tag{17}$$

Here x denotes the elongation in the spring or the displacement of other end which is also the corresponding displacement. To understand the origin of the factor $\frac{1}{2}$ in the above expression, we can think of pulling the spring very slowly. When the displacement of the spring is Δ, the force within the spring is just $k\Delta$. So, to ramp up the displacement from 0 to x, we need to ramp up the force from 0 to kx $(= F)$. The corresponding force-displacement curve is shown in Figure 10.3b. The work done in this quasistatic process will be the area of the shaded triangle $(= \frac{1}{2}F \cdot x)$. However, the final energy stored in an elastic system is independent of the loading path taken in the process: it only depends on the final force/displacement. Thus, even if we apply the force instantly to reach the final stage, the energy stored would still be $\frac{1}{2}Fx$ although the work done by the external agent is Fx. The other half of the energy either gets converted into kinetic energy or dissipates as heat.†

Taking the analogue of spring, we can now note that the energy stored in a deformable body due to the action of force F_1 would be

$$\frac{1}{2}F_1\delta_1 = \frac{1}{2}k_{11}F_1^2 \quad \text{(using (16))}. \tag{18}$$

This is the energy stored due to a single force. We can have multiple forces $F_1, F_2, \ldots F_n$ (with corresponding displacements $\delta_1, \delta_2, \ldots \delta_n$) acting on the deformable body as shown in Figure 10.4.

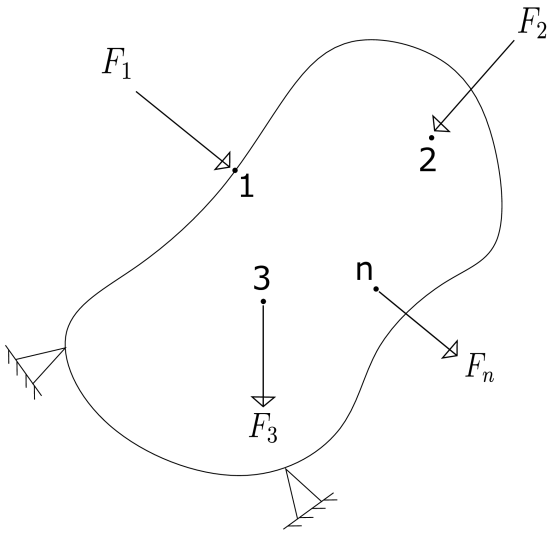


Figure 10.4: Multiple forces applied on a body

The total energy stored would be

$$\sum_{i=1}^{n} \frac{1}{2}F_i\delta_i. \tag{19}$$

In the above, we cannot substitute δ_i as $k_{ii}F_i$ because δ_i contains contributions from multiple forces. It would be the following:

$$\delta_i = \sum_{i=1}^{n} k_{ij}F_j. \tag{20}$$

Plugging this into equation (19), we get

$$\text{Total energy stored} = \sum_i \sum_j \frac{1}{2}k_{ij}F_iF_j. \tag{21}$$

This is the expression of total energy stored in the body in terms of the applied forces and influence coefficients.

10.3 Reciprocal relation: NPTEL link

The reciprocal relation allows us to obtain a relationship between influence coefficients k_{ij} and k_{ji}. Let us work this out through energy considerations. We will consider two different processes. In the first process, think of a body with a force F_1 applied on it at point 1. The energy stored in the body will be

$$E = \frac{1}{2}F_1\delta_1 = \frac{1}{2}F_1\left(k_{11}F_1\right) = \frac{1}{2}k_{11}F_1^2. \tag{22}$$

This action is shown in Figure 10.5 where due to this force, the body deforms from solid black configuration to the dashed red configuration. We call this step 1 where the point 1 gets displaced to 1' during deformation.

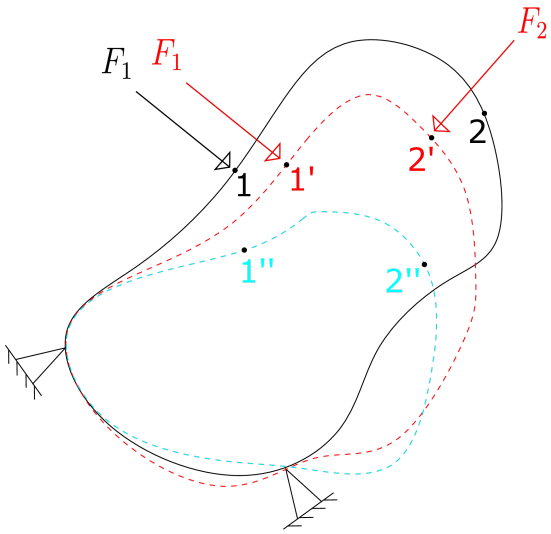


Figure 10.5: Subsequent application of forces F_1 and F_2 on a body.

Then, we apply force F_2 at point 2' in the current configuration which was at point 2 in the original/undeformed body. The body gets further deformed to the dashed blue configuration as shown in Figure 10.5. The point 1' goes to 1" and 2' goes to 2". This is step 2. The energy stored in the body in the final state will be

$$E = \frac{1}{2}k_{11}F_1^2 + \left(\underbrace{F_1(k_{12}F_2)}_{\text{due to } F_1} + \underbrace{\frac{F_2 \cdot k_{22}F_2}{2}}_{\text{due to } F_2}\right)$$
$$= \frac{1}{2}k_{11}F_1^2 + \frac{1}{2}k_{22}F_2^2 + k_{12}F_1F_2 \tag{23}$$

Notice that the contribution to energy due to F_1 during step 2 is not divided by 2. This is because during step two, the full F_1 acts throughout the deformation: the work done will thus simply be the force F_1 times the extra displacement ($1' \rightarrow 1''$). This additional displacement is nothing but the displacement due to F_2, i.e., $k_{12}F_2$. The work done due to the sudden application of force F_2 is the same as the ramped up work. Thus, it is divided by 2. The corresponding displacement in this term represents movement of 2'→2" (and not of 2→ 2") as this part of displacement is the one due to the application of F_2 which can be written as $k_{22}F_2$.
We now think of the second process which is basically the first process only but the order of the two steps are reversed here. This means that we first apply force F_2 and then apply force F_1 in the second step. Due to application of F_2 in the first step, the energy stored in the body will be

$$E = \frac{1}{2}k_{22}F_2^2. \tag{24}$$

In the second step, F_1 is applied. As F_2 is acting fully in this step, the additional energy due to it in step 2 will not be divided by 2. The total energy stored in the body will thus be

$$E = \frac{1}{2}k_{22}F_2^2 + \left(\underbrace{F_2 \cdot k_{21}F_1}_{\text{due to } F_2} + \underbrace{\frac{1}{2}F_1(k_{11}F_1)}_{\text{due to } F_1}\right)$$
$$= \frac{1}{2}k_{22}F_2^2 + \frac{1}{2}k_{11}F_1^2 + K_{21}F_1F_2. \tag{25}$$

Now, in both the processes, the final state is the same with both F_1 and F_2 acting. Thus, the final energy stored in the two processes must be the same. Hence, upon comparing equations (23) and (25), we get

$$k_{12} = k_{21}. \tag{26}$$

Generalizing this, we can write the reciprocal relation as

$$k_{ij} = k_{ji}. \tag{27}$$

10.3.1 Maxwell-Betti-Rayleigh reciprocal theorem: NPTEL link

Consider again a body which is clamped at some points. We think of two situations. In the first situation, multiple forces $F_1, F_2, \ldots F_n$ are applied on it as shown in Figure 10.4. The corresponding displacements for these forces are $\delta_1, \delta_2, \ldots \delta_n$. In the second situation, we apply forces $F_1', F_2', \ldots F_n'$ each at the same location and direction but of different magnitude. We further measure the corresponding displacements $\delta_1', \delta_2', \ldots \delta_n'$. The Maxwell-Betti-Rayleigh reciprocal theorem tells us that

$$\sum_{i=1}^{n} F_i \delta_i' = \sum_{i=1}^{n} F_i' \delta_i. \tag{28}$$

This means that the work done by forces in situation 1 through the corresponding displacements of situation 2 equals the work done by forces in situation 2 through the corresponding displacements of situation 1. Let us try to prove this. Consider the LHS of equation (28):

$$\begin{aligned}\sum_i F_i \delta_i' = \sum_i F_i \sum_j k_{ij} F_j' = {} & F_1(k_{11}F_1' + k_{12}F_2' + \ldots + k_{1n}F_n') \\ & + F_2(k_{21}F_1' + k_{22}F_2' + \ldots + k_{2n}F_n') \\ & + \ldots \\ & + F_n(k_{n1}F_1' + k_{n2}F_2' + \ldots + k_{nn}F_n').\end{aligned} \tag{29}$$

We can rearrange these terms by collecting the first terms within each bracket, second terms within each bracket and so on, i.e.,

$$\begin{aligned}\sum_i F_i \delta_i' = {} & F_1'(k_{11}F_1 + k_{21}F_2 + \ldots + k_{n1}F_n) \\ & + F_2'(k_{12}F_1 + k_{22}F_2 + \ldots + k_{n2}F_n) \\ & + \ldots \\ & + F_n'(k_{1n}F_1 + k_{2n}F_2 + \ldots + k_{nn}F_n) \\ = {} & F_1'(k_{11}F_1 + k_{12}F_2 + \ldots + k_{1n}F_n) \\ & + F_2'(k_{21}F_1 + k_{22}F_2 + \ldots + k_{2n}F_n) \\ & + \ldots \\ & + F_n'(k_{n1}F_1 + k_{n2}F_2 + \ldots + k_{nn}F_n) \quad \text{(using reciprocal relation)} \\ = {} & \sum_i F_i' \delta_i.\end{aligned} \tag{30}$$

This completes the proof of reciprocal theorem.

10.4 Generalized forces and generalized displacements: NPTEL link

Till now, we have only discussed about forces and displacements. The above relations can be extended to moments and rotations as well. Just like the work analog of force is displacement, the work analog of moment is rotation. Consider an arbitrary body on which several forces and moments act (see Figure 10.6).

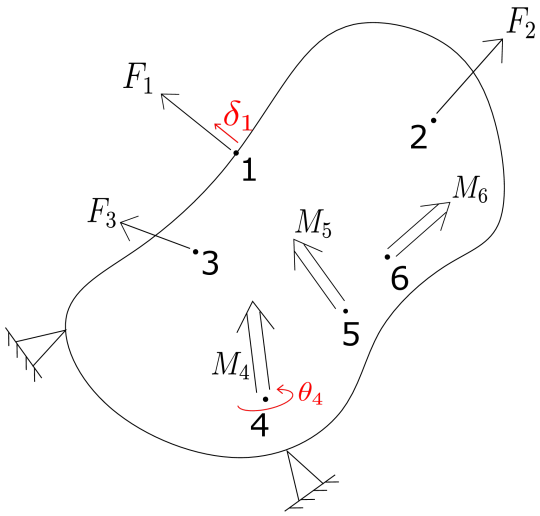


Figure 10.6: A body on which several forces and moments act

We can measure the corresponding displacements and corresponding rotations. A corresponding rotation is the local rotation of the deformable body in the direction of the applied moment at the same point. Moments and rotations can be seen as generalized forces and generalized displacements, respectively. If we think in these terms, all the relations derived earlier do not differentiate between forces and moments or between displacements and rotations. Thus, all the earlier relations will hold for moments and rotations also. For example, the corresponding rotation θ_4 for M_4 will be given by

$$\theta_4 = k_{41}F_1 + k_{42}F_2 + k_{43}F_3 + k_{44}M_4 + k_{45}M_5 + k_{46}M_6. \tag{31}$$

The dimensions of the influence coefficients relating force and rotation and those relating moment and rotation are different though. For example, the dimension of k_{41} is $\frac{\theta}{force}$ or $\frac{1}{force}$ while that of k_{44} is $\frac{\theta}{moment}$ or $\frac{1}{moment}$.[‡] Similarly, the corresponding displacement for force F_1 can be written as

$$\delta_1 = k_{11}F_1 + k_{12}F_2 + k_{13}F_3 + k_{14}M_4 + k_{15}M_5 + k_{16}M_6. \tag{32}$$

The dimension of k_{11} is $\frac{displacement}{force}$ while that of k_{14} is $\frac{displacement}{moment}$. As the dimension of moment equals that of force times displacement, the dimension of k_{14} can also be written as that of $\frac{1}{force}$. This means that the dimensions of k_{14} and k_{41} are the same. So, the reciprocal relation can also be easily extended to mixed combinations (i.e., displacement-moment or rotation-force) and one can write

$$k_{14} = k_{41} \tag{33}$$

as earlier.

10.5 Castigliano's First Theorem: NPTEL link

We have already derived the expression for energy stored in a deformable body as

$$E = \sum_i \frac{1}{2}F_i\delta_i = \sum_i \frac{1}{2}F_i\left(\sum_j k_{ij}F_j\right) = \sum_i\sum_j \frac{1}{2}k_{ij}F_iF_j. \tag{34}$$

Let us take the first derivative of this expression with respect to the kth generalized force:

$$\begin{aligned}\frac{\partial E}{\partial F_k} &= \sum_i\sum_j \frac{1}{2}k_{ij}\left(\frac{\partial F_i}{\partial F_k}F_j + F_i\frac{\partial F_j}{\partial F_k}\right)\\ &= \sum_i\sum_j \frac{1}{2}k_{ij}(\delta_{ik}F_j + F_i\delta_{jk}) = \frac{1}{2}\left(\sum_j k_{kj}F_j + \sum_i k_{ik}F_i\right).\end{aligned} \tag{35}$$

Upon further replacing the summation index j with i, we get

$$\frac{\partial E}{\partial F_k} = \frac{1}{2}\sum_i (k_{ki}F_i + k_{ik}F_i) = \sum_i k_{ki}F_i \quad \text{(using reciprocal relation)}$$
$$= \delta_k. \tag{36}$$

We have thus derived that the corresponding displacement δ_k is the derivative of total energy with respect to the corresponding force F_k. This is called the Castigliano's first theorem. If we can write the total energy stored in a body in terms of the applied forces/moments, we can apply this theorem to find the corresponding displacement/rotation. We need not solve any differential equation. Of course, we need to know all the influence coefficients.

10.6 Application of Castigliano's theorem for solving beam problems: NPTEL link

Writing the stored energy in terms of externally applied generalized forces is crucial if we want to use the Castigliano's first theorem. Let us see how this can be achieved. Think of a general three-dimensional body which is clamped at some points as shown in Figure 10.7a.

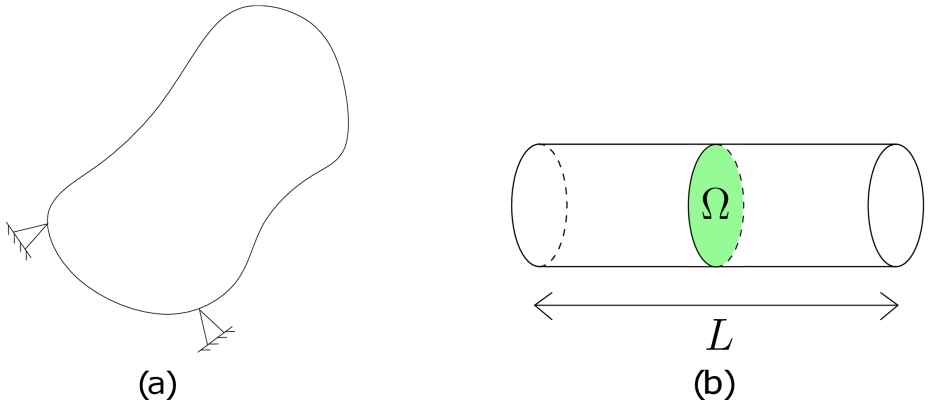


Figure 10.7: (a) A general three-dimensional body clamped at two points (b) A beam having cross-section Ω and length L

We want to know the energy stored in this body. For linear stress-strain relation, the energy stored becomes

$$E = \sum_i \sum_j \iiint_V \frac{1}{2}\sigma_{ij}\epsilon_{ij} dV. \tag{37}$$

Let us simplify this further in the context of beams which we will be mostly dealing with. Consider the beam shown in Figure 10.7b having cross-section Ω and length L. The equation (37) can be rewritten as follows for a beam:

$$E = \int_0^L ds \left(\sum_i \sum_j \iint_\Omega \frac{1}{2}\sigma_{ij}\epsilon_{ij}\, d\Omega \right) \tag{38}$$

We will call the integral in the parentheses above the *cross-sectional strain energy* as it is obtained by integration of the three-dimensional strain energy density over the beam's cross-section. It is further integrated over the length of the beam to obtain the total energy stored in a beam. A beam can deform in several modes such as bending, twisting, stretching and shearing. For each of these deformation modes, the energy stored in the beam's cross-section has different mathematical form. Let's discuss them one by one.

Stretching energy

Consider a beam being stretched by the application of an axial load P as shown in Figure 10.8.

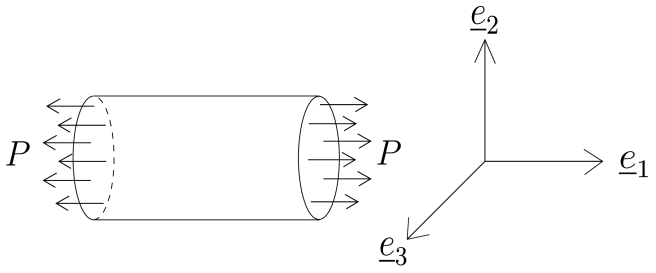


Figure 10.8: A beam being stretched by the application of load P on both its ends

The total load P can be assumed to be uniformly distributed over the cross-section Ω. To find the energy stored in the cross-section, we need to first find the stress and strain components. We can recall from earlier discussion that in case of axial extension of a beam, the cross-section is allowed to relax freely. So, the stress matrix has only one non-zero stress component, i.e.,

$$[\underline{\underline{\sigma}}] = \begin{bmatrix} \sigma_{11} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{39}$$

Using three-dimensional Hooke's Law, we can then write

$$\epsilon_{11} = \frac{1}{E}(\sigma_{11} - \nu(\sigma_{22} + \sigma_{33})) = \frac{\sigma_{11}}{E}. \tag{40}$$

The cross-sectional energy will thus become

$$\sum_i \sum_j \iint_\Omega \frac{1}{2}\sigma_{ij}\epsilon_{ij} d\Omega = \iint_\Omega \frac{1}{2}\sigma_{11}\epsilon_{11} d\Omega = \iint_\Omega \frac{\sigma_{11}^2}{2E} d\Omega = \iint_\Omega \frac{P^2}{2EA^2} d\Omega. \tag{41}$$

In the last step, we assumed that the axial load P is uniformly distributed over the cross-section. Upon final integration over the cross-section, we get

$$E_{stretching} = \frac{P^2}{2EA}. \tag{42}$$

This can be further integrated over the length of the beam to get the total energy due to stretching of a beam.

Bending energy

Figure 10.9 shows a beam which bends due to the application of bending moment M.

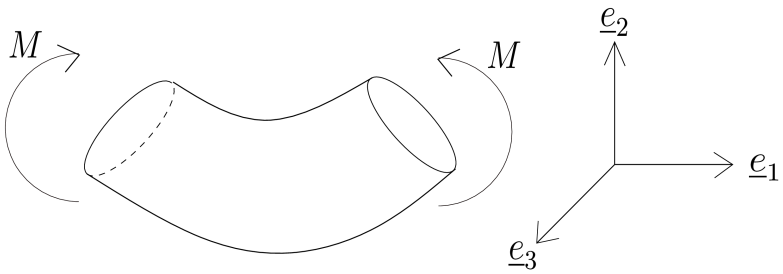


Figure 10.9: A beam under pure bending.

We can recall that the state of stress for such a case is given by

$$[\underline{\underline{\sigma}}] = \begin{bmatrix} \sigma_{11} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{43}$$

This is because the cross-section again relaxes freely in the transverse direction in case of pure bending. We had also derived that

$$\kappa = \frac{M_z}{EI_{zz}},\ \epsilon_{11} = -\kappa y = \frac{-M_z y}{EI_{zz}},\ \sigma_{11} = -E\kappa y = \frac{-M_z y}{I_{zz}}. \tag{44}$$

Here κ denotes the bending curvature and y denotes distance from the neutral axis. Thus, the cross-sectional energy will be

$$\sum_i \sum_j \iint_\Omega \frac{1}{2}\sigma_{ij}\epsilon_{ij} d\Omega = \iint_\Omega \frac{1}{2}\sigma_{11}\epsilon_{11} d\Omega = \iint_\Omega \frac{M_z^2}{2EI_{zz}^2} y^2 d\Omega. \tag{45}$$

The final integration over the cross-section yields

$$E_{bending} = \frac{M_z^2}{2EI_{zz}}. \tag{46}$$

If we compare this form of bending energy with the extensional/stretching energy form (42), we can see that the form of both the energies is exactly the same: the axial force P is replaced by bending moment M while the stretching stiffness EA is replaced by bending stiffness EI_{zz}.

Torsional energy

In case of torsion, the state of stress in cylindrical coordinate system is

$$\left[\underline{\underline{\sigma}}\right] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \tau_{\theta z} \\ 0 & \tau_{\theta z} & 0 \end{bmatrix}. \tag{47}$$

We had also derived earlier that

$$\kappa = \frac{T}{GJ},\ \gamma_{\theta z} = \kappa r = \frac{Tr}{GJ},\ \tau_{\theta z} = G\kappa r = \frac{Tr}{J}. \tag{48}$$

Here κ denotes twist, r denotes radial coordinate relative to the cross-section's centroid and T denotes the internal torque or twisting moment. Thus, the cross-sectional energy will be

$$\sum_i \sum_j \iint_\Omega \frac{1}{2}\sigma_{ij}\epsilon_{ij} d\Omega = \iint_\Omega \frac{1}{2}[\tau_{\theta z}\epsilon_{\theta z} + \tau_{z\theta}\epsilon_{z\theta}] d\Omega = \iint_\Omega \frac{1}{2}\tau_{\theta z}\gamma_{\theta z} d\Omega = \iint_\Omega \frac{T^2}{2GJ^2} r^2 d\Omega.$$

The final integration over the cross-section yields

$$E_{twisting} = \frac{T^2}{2GJ}. \tag{49}$$

Shear energy due to transverse load

We had learnt earlier that the presence of transverse load leads to non-uniform bending of beams which also generates transverse shear stress in beams - this shear stress points in the direction of applied transverse load and is different from the one due to torsion.[§] Proceeding along the similar lines as earlier, the total shear energy due to transverse load turns out to be

$$E_{shearing} = \frac{V^2}{2k_s GA}. \tag{50}$$

Here V represents shear force, GA shearing stiffness and k_s shear correction factor.

Total energy stored in a beam due to general loading: NPTEL link

When we apply general load on a beam, force and moment both act on the cross-section (see Figure 10.10).

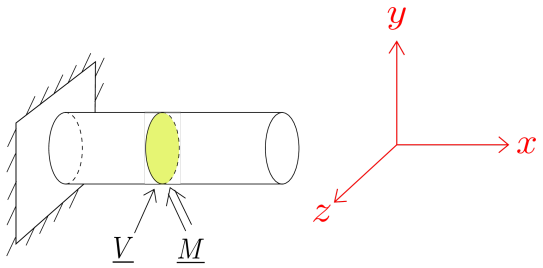


Figure 10.10: A beam clamped at one end and subjected to general loading in the cross-section

The internal force $\underline{V}$ in the cross-section can be decomposed into three components as

$$[\underline{V}] = [V_x \quad V_y \quad V_z]^T. \tag{51}$$

The component along the axis V_x is axial force P. The other components are shear forces. We can similarly resolve the internal moment into three components as

$$[\underline{M}] = [M_x \quad M_y \quad M_z]^T. \tag{52}$$

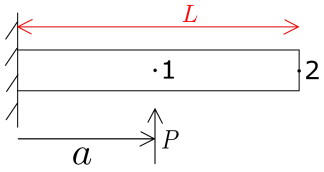
The component along the axis M_x is the torque T. The other two components are bending moments. We derived energy due to each of these six components separately assuming only one of them was present. For the general case, can we sum individual energies? It turns out we cannot since individual energies are quadratic in applied generalized forces. However, if we resolve internal force and moment along principal directions, one can show that the total cross-sectional energy will indeed be simply the summation of energies of individual modes, i.e.,

$$E = \int_0^L \left[\frac{P(x)^2}{2EA} + \frac{M_y(x)^2}{2EI_{zz}} + \frac{M_z(x)^2}{2EI_{zz}} + \frac{T(x)^2}{2GJ} + \frac{V_y(x)^2}{2kGA} + \frac{V_z(x)^2}{2kGA} \right] dx. \tag{53}$$

The first three terms here are due to normal strain whereas the last three terms are due to shear strain.

10.7 Solved examples

Q1. Think of a beam clamped at one end and acted upon by a force P at a distance a from the clamped end. Find tip rotation of the beam. Suppose a moment is then applied at the free end - obtain the beam's deflection at point 1. Use this to verify reciprocal relation. (NPTEL link)



Solution: Let us first obtain the beam's tip rotation, i.e., $\theta(L)$. Let us solve it using Euler-Bernouli beam theory. We denote the point at which P is acting as point 1 and the free end as point 2. According to EBT:

$$EI\frac{d^2y}{dx^2} = M(x). \tag{54}$$

The bending moment profile can be found by cutting a section in the beam and analyzing one of its portions. It turns out to be

$$\begin{aligned} M &= 0 \qquad\quad (x > a) \\ &= F_1(a - x) \quad (x < a) \end{aligned} \tag{55}$$

Here we are writing the force P as F_1 since it is acting at point 1. Substituting this moment profile in equation (54) gives:

$$\begin{aligned} EI\frac{d^2y}{dx^2} &= 0 \qquad\quad (x > a) \\ &= F_1(a - x) \quad (x < a) \end{aligned} \tag{56}$$

Integrating the expression for $x < a$ twice gives

$$EIy = F_1 a\frac{x^2}{2} - F_1\frac{x^3}{6} + C_1 x + C_2. \tag{57}$$

The boundary condition at the clamped end is

$$y(0) = 0, \quad \theta(0) = \frac{dy}{dx}(0) = 0 \tag{58}$$

which when applied yields

$$C_1 = C_2 = 0. \tag{59}$$

Thus, we get

$$y = \frac{F_1}{EI}\left(a\frac{x^2}{2} - \frac{x^3}{6}\right) \tag{60}$$

and its derivative gives cross-sectional rotation θ, i.e.,

$$\theta = \frac{dy}{dx} = \frac{F_1}{EI}\left(ax - \frac{x^2}{2}\right). \tag{61}$$

The above two relations hold for $x \leq a$ only. In order to obtain expression for $x > a$, we set $M = 0$ in EBT theory, i.e.,

$$\frac{d^2y}{dx^2} = 0 \Rightarrow \frac{d\theta}{dx} = 0. \tag{62}$$

Thus, θ is constant in the segment from $x = a$ to $x = L$ which implies

$$\theta(L) = \theta(a) = \frac{F_1 a^2}{2EI}. \tag{63}$$

The second equality is obtained after enforcing continuity of rotation at $x = a$ and further evaluating equation (61) at $x = a$. The above equation gives us influence coefficient relating θ at point 2 with force at point 1, i.e.,

$$\theta_2 = \underbrace{\frac{a^2}{2EI}}_{k_{21}} F_1 \tag{64}$$

We can also find k_{12} to check whether it comes out to be equal to k_{21}. To obtain k_{12}, we apply a bending moment M_2 at point 2 and measure the deflection at point 1 (see Figure 10.11).

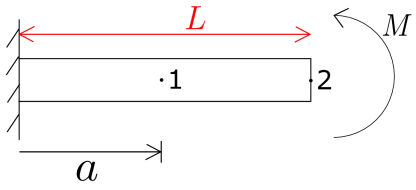


Figure 10.11: A bending moment M applied at point 2 instead of force P being applied at point 1

In this case, bending moment will be constant all along the length of the beam and equal to the applied moment M_2. Substituting this in equation (54), we get

$$EI\frac{d^2y}{dx^2} = M(x) = M_2 \Rightarrow y = \frac{1}{EI}\left[\frac{M_2\,x^2}{2} + C_1x + C_2\right]. \tag{65}$$

The boundary conditions at the clamped end are the same as before (see (58)), which again yield $C_1 = C_2 = 0$. Thus, we get

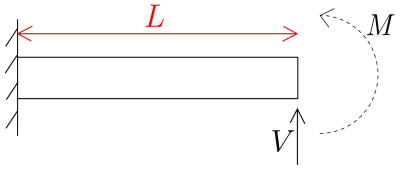
$$y = \frac{M_2\,x^2}{2EI} \tag{66}$$

using which the deflection at point 1 turns out to be

$$y_1 = y(a) = \underbrace{\frac{a^2}{2EI}}_{k_{12}} M_2. \tag{67}$$

We can see that k_{12} is the same as k_{21}. Thus, we are able to verify the reciprocal relation. As we can see, the second problem of pure bending is much easier to solve than the first problem of finding the tip rotation. If we are asked to solve for tip rotation, we actually need k_{21}. But if we can find k_{12} instead in a simple way (as we saw above), we can then use reciprocal relation to set it equal to k_{21}. Thus, reciprocal relation turns out to be very useful in obtaining solution to a difficult problem by solving a simpler problem.

Q2. Consider a beam clamped at one end and a transverse load is applied at the other end. Can we obtain the beam's tip rotation using energy methods? (NPTEL link)



Solution: We want to find the rotation of the cross-section at the tip of the beam using energy method. The corresponding displacement of shear force is tip deflection. In order to find the rotation at the tip, a bending moment would also be needed to be applied at the tip: the unknown $\theta(L)$ is the corresponding displacement of bending moment $M(L)$. So, the steps needed to be followed are:

(1) Apply a dummy bending moment M at the tip.

(2) Obtain the energy of entire beam in terms of (V, M).

(3) Apply Castigliano's first theorem, i.e., obtain the partial derivative $\frac{\partial E}{\partial M}$ which equals θ_L.

(4) Finally, set $M = 0$ in the expression of θ_L since no bending moment acts at the tip in the original problem.

To obtain the beam's energy, we first need shear force and bending moment profiles along the length of the beam. If we cut a section in the beam at a distance x from the clamped end and balance forces and moments

in the right part of the cut beam, we will find that

$$V_y(x) = V, \quad M_z(x) = M + V(L-x). \tag{68}$$

The axial force P, shear force V_z, torque T and bending moment M_y are all zero here. Using these in equation (53), we get

$$E = \int_0^L \left[\frac{V_y^2}{2kGA} + \frac{M_z^2}{2EI_{zz}}\right] dx = \int_0^L \left[\frac{V^2}{2kGA} + \frac{(M+V(L-x))^2}{2EI_{zz}}\right] dx. \tag{69}$$

We have thus been able to write the energy stored in the beam in terms of the externally applied loads (V and M). To obtain θ_L, we now apply Castigliano's first theorem, i.e.,

$$\theta_L = \frac{\partial E}{\partial M} = \int_0^L \left[\frac{2V\frac{\partial V}{\partial M}}{2kGA} + \frac{2(M+V(L-x))\frac{\partial}{\partial M}(M+V(L-x))}{2EI_{zz}}\right] dx. \tag{70}$$

As V and M are independent quantities, $\frac{\partial V}{\partial M}$ would be zero which leads to

$$\theta_L = \int_0^L \frac{2(M+V(L-x))}{2EI_{zz}} dx \tag{71}$$

We finally set $M = 0$ to obtain

$$\theta_L = \int_0^L \frac{2(0+V(L-x))}{2EI_{zz}} dx = \frac{V}{EI_{zz}} \int_0^L (L-x) dx = \frac{V}{EI_{zz}} \left(Lx - \frac{x^2}{2}\right)\Big|_0^L = \frac{VL^2}{2EI_{zz}}.$$

For an exhaustive set of solved examples and unsolved objective and subjective type questions, please refer to the printed book: click here

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*If we consider large deformation problem, then stress-strain relation and strain-displacement relation become nonlinear.

†In the quasistatic process, the work done by the external agent exactly equals the energy stored in the spring since no part of the work done is dissipated.

‡Note that rotation is dimensionless

§In torsion of beams, total shear force due to shear stress $\tau_{\theta z}$ is zero whereas the resultant shear force due to shear stress under transverse loading is obviously non-zero.